

Computer Networks

Lecture 01: Introduction to Computer Networks

Department of Computer Science

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1 Introduction to Computer Networks

In today's world, computers and other digital devices rarely work in complete isolation. A computer may need to communicate with another computer, access a web server, send an email, download a file, communicate with a smartphone, access a cloud service, or share a printer with other devices.

The technology that makes this communication possible is called **computer networking**.

A computer network allows two or more computing devices to communicate and exchange information through a communication medium by following a common set of rules called protocols.

For example, when a student opens a web browser and visits a website, many networking operations take place in the background. The student's computer or smartphone sends requests through a network, routers forward packets toward the destination, the server receives the request, and the server sends data back to the user's device.

Therefore, a computer network is not simply a collection of connected computers. It is a complete communication system involving devices, communication media, addressing mechanisms, protocols, and different functions that make communication possible and useful.

2 Definition of a Computer Network

2.1 Basic Definition

A **computer network** is a collection of two or more interconnected computing devices that communicate with each other and exchange data or share resources through a wired or wireless communication medium using predefined communication protocols.

In simple words:

A computer network is a system in which devices are connected so that they can communicate, exchange information, and share resources.

The devices in a network may include:

- Desktop computers
- Laptop computers
- Smartphones
- Tablets
- Servers
- Printers

- Routers
- Switches
- Access points
- Network-attached storage devices
- Internet of Things (IoT) devices

A network can be very small, such as two computers connected directly, or extremely large, such as the Internet, which connects networks around the world.

3 Basic Elements of Network Communication

For communication to take place in a computer network, several important elements are involved.

The most fundamental elements are:

1. Sender
2. Receiver
3. Data or Message
4. Communication Medium
5. Protocol

These elements work together to make communication possible.

3.1 Sender

The **sender** is the device or system that originates the data.

In other words, the sender is the source of a communication.

Examples include:

- A laptop sending an email
- A smartphone sending a WhatsApp message
- A computer uploading a file to a server
- A web browser sending a request to a web server

For example, if Computer A sends a file to Computer B, Computer A is the sender.

Computer A (Sender) $\longrightarrow$ Computer B

3.2 Receiver

The **receiver** is the device or system that receives the transmitted data.

For example, if a student sends a message from a smartphone to another student's smartphone, the first smartphone is the sender and the second smartphone is the receiver.

The roles of sender and receiver are not necessarily permanent. In a two-way communication system, the same device can act as a sender at one moment and as a receiver at another moment.

3.3 Data or Message

The information being communicated is called the **data** or **message**.

The data may contain:

- Text
- Numbers
- Images
- Audio
- Video
- Documents
- Software files
- Web pages
- Control information

Although users see different types of information, computers ultimately represent digital information using binary values, mainly 0 and 1.

For example, the text:

HELLO

is internally represented using binary data.

3.4 Communication Medium

The **communication medium** is the path through which data travels from the sender toward the receiver.

Communication media are generally divided into two major categories:

1. Wired media
2. Wireless media

3.4.1 Wired Communication

In wired communication, a physical medium is used.

Examples include:

- Twisted-pair cable
- Coaxial cable
- Fiber-optic cable

For example, an Ethernet cable can connect a computer to a network switch or router.

3.4.2 Wireless Communication

In wireless communication, there is no physical cable connecting the communicating devices through the air interface.

Wireless communication commonly uses electromagnetic waves.

Examples include:

- Wi-Fi
- Bluetooth
- Cellular networks
- Microwave communication
- Satellite communication

A smartphone connected to Wi-Fi is communicating wirelessly with the wireless access point.

3.5 Protocol

A **protocol** is a set of rules that defines how devices communicate and exchange data over a network.

Without common rules, devices would not know:

- How data should be formatted
- Where data should be sent
- How the receiver should interpret the data
- How errors should be handled

- How communication should begin and end

A simple real-world analogy is human conversation.

If two people want to communicate, they need some common understanding of language and communication rules. Similarly, computers need networking protocols.

Examples of networking protocols include:

- HTTP
- HTTPS
- TCP
- UDP
- IP
- DNS
- Ethernet
- Wi-Fi standards

For example, HTTP is used for communication between web browsers and web servers.

Modern computer networks rely heavily on protocol suites such as TCP/IP. Protocols provide the rules that allow different systems and devices to communicate even when they are made by different manufacturers.

4 How Communication Takes Place in a Network

A simplified communication process can be represented as:

Sender → Communication Medium → Receiver

However, real computer networks are usually more complicated.

For example:

Laptop → Wi-Fi Access Point → Router → Internet → Server

The sender prepares data, networking protocols organize the communication, the data is transmitted through a medium, intermediate network devices may forward it, and eventually the receiver obtains the data.

5 Digital Data and Physical Signals

One important concept that students should understand from the beginning is the difference between **data** and the **signal used to carry that data**.

Computers normally process digital information.

Digital information is commonly represented using binary values:

0 and 1

However, a 0 or 1 cannot simply travel through the air or through a cable as an abstract mathematical value.

The binary information must be represented by a physical signal.

The signal may involve:

- Electrical changes
- Light pulses
- Radio waves
- Changes in amplitude
- Changes in frequency
- Changes in phase

Therefore, we must distinguish between:

Digital Data and Physical Signal

5.1 Is Wireless Data Analog or Digital?

A common misunderstanding is:

“Wireless communication is analog and wired communication is digital.”

This statement is incorrect.

Wireless networking can carry **digital data**.

For example, when a smartphone sends the binary data of a photograph through Wi-Fi, the information being transmitted is digital.

However, the physical radio signal used to carry those bits is an electromagnetic waveform. The digital information is represented by controlled changes in properties of the wireless signal.

Therefore:

Digital Data $\longrightarrow$ Encoding/Modulation $\longrightarrow$ Physical Signal

At the receiver:

Physical Signal $\longrightarrow$ Demodulation/Decoding $\longrightarrow$ Digital Data

For example, Wi-Fi transmits digital information using radio-frequency signals.

Similarly, a wired Ethernet network carries digital information over physical signals in a cable.

5.2 What Does “Analog” Mean?

An **analog signal** varies continuously with time.

For example, a continuously varying electrical or electromagnetic waveform can represent information.

A simplified analog waveform may look conceptually like a smooth wave:

Continuous variation of signal over time

Digital signaling, on the other hand, represents information using discrete signal states or symbols.

A simplified example is:

1 0 1 1 0 0 1

However, it is important to understand that even digital communication ultimately uses a physical waveform. The terms “digital” and “analog” describe how information is represented and signaled, not whether a cable or wireless medium is used.

6 Types of Computer Networks

Computer networks can be classified in several ways.

One common classification is based on the geographical area covered by the network.

Important types include:

1. Personal Area Network (PAN)
2. Local Area Network (LAN)
3. Metropolitan Area Network (MAN)
4. Wide Area Network (WAN)

6.1 Personal Area Network (PAN)

A **Personal Area Network (PAN)** is a small network around an individual person. The coverage area is usually very small.

Examples include:

- Smartphone connected to wireless earbuds
- Smartphone connected to a smartwatch
- Laptop connected to a Bluetooth mouse
- Personal devices connected through Bluetooth

Bluetooth is a common technology used for PAN communication.

6.2 Local Area Network (LAN)

A **Local Area Network (LAN)** connects devices within a relatively limited geographical area.

Examples include:

- Computer lab
- University department
- Home network
- Office network
- School network

A LAN may use Ethernet cables, Wi-Fi, or a combination of wired and wireless technologies.

For example, computers in a university computer laboratory may be connected to switches using Ethernet cables.

6.3 Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** covers a larger area than a typical LAN and may span a city or metropolitan region.

For example, a network connecting multiple buildings, campuses, or facilities across a city can be considered a MAN.

The distinction between LAN and MAN is mainly related to scale and geographical coverage.

6.4 Wide Area Network (WAN)

A **Wide Area Network (WAN)** covers a large geographical area.

It may connect:

- Cities
- Provinces
- Countries
- Continents

The Internet is the most famous example of a very large interconnected network.

Organizations may also create private WANs to connect offices located in different cities or countries.

7 Functionality of Computer Networks

A computer network performs many different functions.

The basic purpose is communication and resource sharing, but reliable networking requires many supporting mechanisms.

The term **network functionality** refers to the different tasks and services performed by networking systems and protocols to make communication possible, reliable, efficient, secure, and manageable.

Examples include:

- Error control
- Flow control
- Multiplexing
- Demultiplexing
- Addressing
- Routing
- Forwarding
- Data segmentation and reassembly
- Access control
- Synchronization
- Security

- Encryption
- Recovery and resynchronization

It is important to understand that not every networking protocol performs all of these functions.

Different functions are performed at different layers and by different protocols.

For this introductory lecture, we will divide some important functions into:

1. Mandatory or fundamental communication functions
2. Optional or service-enhancing functions

Important Note: The word “mandatory” here means that these functions or mechanisms are fundamental to successful network communication in the appropriate networking context. It does not mean that every single network protocol implements every function in exactly the same way.

There are many networking functionalities. In this lecture, we discuss only some of the most important ones.

8 Mandatory or Fundamental Network Functions

The following functions are fundamental concepts that students must understand because they appear repeatedly throughout computer networking.

8.1 Error Control

Error control refers to the mechanisms used to detect and, where required, recover from errors that occur during data transmission.

When data travels through a network, it may be affected by different problems.

For example:

- Electrical interference
- Radio interference
- Noise
- Signal degradation
- Packet loss
- Damaged frames or packets

As a result, the data received by a receiver may not always be exactly the same as the data sent by the sender.

Error control attempts to maintain data correctness.

8.1.1 Example of Error Control

Suppose a sender transmits:

10110010

Because of transmission problems, the receiver may obtain:

10100010

One bit has changed.

An error-control mechanism can help detect that something went wrong.

Depending on the protocol and technique being used, the system may:

1. Detect the error
2. Discard the damaged data
3. Request retransmission
4. Correct the error automatically

Common concepts associated with error control include:

- Error detection
- Error correction
- Acknowledgement
- Retransmission
- Checksum
- Cyclic Redundancy Check (CRC)

8.1.2 Error Detection

Error detection means determining whether transmitted data has been corrupted.

For example, a checksum or CRC may be calculated from transmitted data.

The receiver performs a corresponding calculation.

If the calculated value does not match the received value, the receiver can determine that an error has occurred.

8.1.3 Retransmission

When a protocol supports retransmission, damaged or lost data can be transmitted again.
For example:

$$\text{Sender} \rightarrow \text{Data} \rightarrow \text{Receiver}$$

If the receiver detects a problem:

$$\text{Receiver} \rightarrow \text{Request/Negative Response} \rightarrow \text{Sender}$$

The sender may then transmit the required data again.

TCP is an important example of a protocol that provides reliability mechanisms including retransmission.

8.2 Flow Control

Flow control is the mechanism used to control the rate at which a sender transmits data so that the receiver is not overwhelmed.

Consider a sender that can transmit data very quickly while the receiver can process data more slowly.

Suppose:

$$\text{Sender speed} = 1000 \text{ units/second}$$

while:

$$\text{Receiver processing speed} = 500 \text{ units/second}$$

If the sender continuously transmits at its maximum rate, the receiver may not have enough buffer space to store the incoming data.

This can result in data loss.

Therefore, flow control helps match the sender's transmission rate with the receiver's ability to accept and process data.

IBM describes flow control as a mechanism that manages the rate of transmission between communicating devices so that the receiver is not overwhelmed.

8.2.1 Simple Example of Flow Control

Imagine that a teacher is giving papers to students.

If the teacher gives papers faster than students can organize them, papers may pile up.

The teacher therefore slows down when necessary.

Similarly:

Fast Sender $\longrightarrow$ Flow Control $\longrightarrow$ Slower Receiver

Flow control is therefore concerned with the relationship between the sender and receiver.

8.3 Multiplexing

Multiplexing is a technique in which multiple independent data streams share a common communication channel or connection.

The device or mechanism performing this operation is commonly called a **multiplexer**, or **MUX**.

Instead of requiring a separate physical communication path for every data stream, multiple streams can share one communication resource.

Conceptually:

Data Stream 1 Data Stream 2 Data Stream 3

are combined:

↓

MUX

↓

Shared Communication Channel

This improves the utilization of communication resources.

8.3.1 Real-World Example of Multiplexing

Suppose a computer is simultaneously:

- Browsing a website
- Downloading a file
- Using an online messaging application
- Watching an online video

All of these applications can communicate through the same network interface and connection to the network.

The networking system uses identifiers such as port numbers and protocol information to distinguish different communication streams.

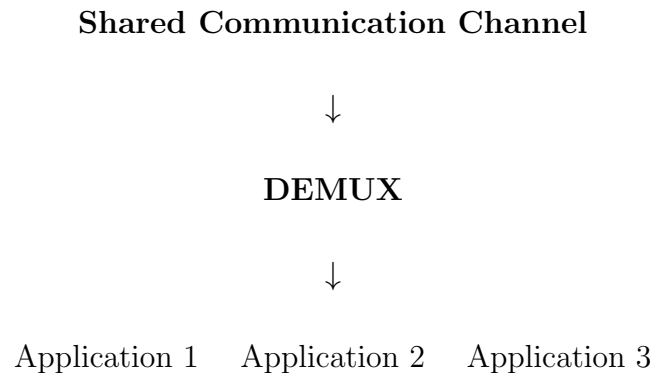
8.4 Demultiplexing

Demultiplexing is the reverse process of multiplexing.

The receiver receives combined network traffic and determines which portion of the traffic belongs to which application or communication process.

The device or mechanism performing this operation can be called a **demultiplexer**, or **DEMUX**.

Conceptually:



For example, the transport layer uses port numbers to help deliver received data to the correct application.

IBM identifies multiplexing as a transport-layer function that allows multiple applications to use the same network connection.

8.5 Addressing

Addressing is the process of identifying communicating devices, interfaces, applications, or services so that data can be delivered to the correct destination.

Imagine sending a letter.

The postal system needs an address to determine where the letter should go.

Similarly, computer networks require addressing information.

Different networking layers use different types of addresses.

Examples include:

- MAC addresses
- IP addresses
- Port numbers

An IP address can identify a network-layer destination, while a port number can identify a particular service or application endpoint.

For example:

IP Address = “Which device/network interface?”

and:

Port Number = “Which application/service?”

This is a simplified conceptual explanation, but it is very useful for beginners.

8.6 Routing and Forwarding

Another fundamental network function is determining where data should go.

Routing is the process of determining a path through a network toward a destination.

Forwarding is the actual process of sending a received packet toward the next appropriate destination or next hop.

For example:

Computer A → Router 1 → Router 2 → Router 3 → Server

The routers examine addressing information and use routing information to determine where packets should be forwarded.

Routing is especially important in large networks such as the Internet.

8.7 Segmentation and Reassembly

Large amounts of data are often divided into smaller units before transmission.

This process is called **segmentation**.

For example, a large file may be divided into many smaller pieces.

Conceptually:

Large Data

↓

Segment 1 Segment 2 Segment 3 Segment 4

The receiver can then process the pieces and reconstruct the original data.

This is important because networks operate using manageable units of data rather than always transmitting an entire large file as one enormous unit.

8.8 Synchronization

Synchronization helps communicating devices coordinate the timing and order of data exchange.

The sender and receiver must have sufficient understanding of where data begins, where it ends, and how pieces of communication relate to one another.

Synchronization becomes particularly important when data is transmitted as a continuous stream or when a communication session must be restored after a disruption.

9 Optional or Service-Enhancing Network Functions

Not every network requires every advanced functionality.

Some functions are provided when a particular application or network requires additional services such as confidentiality, recovery, or performance improvement.

Important examples include:

1. Encryption and cryptography
2. Checkpointing and recovery
3. Compression
4. Authentication
5. Quality of Service (QoS)

9.1 Encryption and Cryptography

Encryption is the process of transforming readable data, called plaintext, into an encoded form called ciphertext so that unauthorized parties cannot easily understand the information.

The original data can be represented as:

Plaintext

After encryption:

Ciphertext

The receiver uses an appropriate decryption process to recover the original information.

Conceptually:

Plaintext $\rightarrow$ Encryption $\rightarrow$ Ciphertext

and:

Ciphertext $\rightarrow$ Decryption $\rightarrow$ Plaintext

Cryptography is a broader field that includes techniques used for:

- Confidentiality
- Integrity
- Authentication
- Secure communication

For example, HTTPS uses cryptographic mechanisms to protect web communication.

Encryption is not required for every form of communication, but it is extremely important when sensitive information must be protected.

9.2 Checkpointing and Recovery

Checkpointing is a mechanism in which a system records a known point of progress so that communication or processing can resume from that point after an interruption.

Consider a large file download.

Suppose a user has downloaded:

70%

of a large file and then the Internet connection is interrupted.

A system with suitable resume capability may continue from approximately the point already completed instead of starting the entire download again.

Conceptually:

Start $\rightarrow$ 25% $\rightarrow$ 50% $\rightarrow$ 75% $\rightarrow$ 100%

If a failure occurs at 75%, the system may use the saved progress information to resume from a suitable point.

This concept is related to synchronization and recovery mechanisms in networking and distributed systems.

9.3 Compression

Data compression reduces the amount of data that needs to be transmitted or stored. For example, if a file can be compressed from:

$$100 \text{ MB} \rightarrow 60 \text{ MB}$$

then less data may need to be transferred.

Compression can reduce bandwidth requirements and sometimes improve transmission performance.

However, compression itself requires processing and is not suitable or necessary in every situation.

9.4 Authentication

Authentication is the process of verifying the identity of a user, device, or system.

For example, when a user logs into an online service using a username and password, the system attempts to determine whether the user is actually authorized to use that account.

Authentication can use:

- Passwords
- Digital certificates
- Cryptographic keys
- One-time passwords
- Biometrics

Authentication is an important security function in modern networks.

9.5 Quality of Service (QoS)

Quality of Service (QoS) refers to techniques used to manage network traffic so that important applications can receive appropriate network resources and performance.

Not all network traffic has the same requirements.

For example:

- Voice communication is sensitive to delay.
- Video conferencing requires reasonably consistent delivery.
- File downloads can often tolerate more delay.

QoS mechanisms can be used to prioritize certain traffic or manage network resources.

10 Communication in Computer Networks

The word **communication** simply means the exchange of information between communicating entities.

In computer networking, communication means the exchange of data between computing devices or network systems through a communication medium according to defined protocols.

For example:

Laptop $\longleftrightarrow$ Router

or:

Client $\longleftrightarrow$ Server

The communication may occur in one direction or in both directions.

Based on the direction of data flow, communication can be classified as:

1. Simplex communication
2. Half-duplex communication
3. Full-duplex communication

11 Simplex Communication

Simplex communication is communication in which data flows in only one direction.

One device acts as the sender and the other device acts as the receiver.

There is no normal reverse communication path for data.

Conceptually:

Sender $\longrightarrow$ Receiver

but not:

Receiver $\longrightarrow$ Sender

11.1 Example of Simplex Communication

A traditional example is a broadcast system.

For example:

Radio Station $\longrightarrow$ Listeners

The station broadcasts information and listeners receive it.

Another simplified example is a keyboard sending input to a computer, although modern computer interfaces involve more complex mechanisms and feedback.

The important concept is:

In simplex communication, communication is essentially one-way.

12 Half-Duplex Communication

Half-duplex communication allows communication in both directions, but not at the same time.

Therefore, both devices can transmit and receive, but they must take turns.

Conceptually:

Device A $\longrightarrow$ Device B

then:

Device B $\longrightarrow$ Device A

The two directions cannot normally operate simultaneously.

12.1 Example: Walkie-Talkie

A walkie-talkie is a familiar example.

One person presses the transmit button and speaks.

During that time, the other person listens.

Then the first person stops transmitting, and the second person can transmit.

Therefore:

A talks $\rightarrow$ B listens

then:

B talks $\rightarrow$ A listens

This is half-duplex communication.

The main advantage is that both directions can use the same communication resource, but simultaneous transmission is not possible.

13 Full-Duplex Communication

Full-duplex communication allows both devices to transmit and receive at the same time.

Conceptually:

$$\text{Device A} \rightleftarrows \text{Device B}$$

Both directions can operate simultaneously.

13.1 Example: Telephone Conversation

A telephone conversation is a common example.

Person A can speak while Person B is listening, and Person B can also speak while Person A is listening.

Both sides can potentially speak and listen at the same time.

Therefore:

$$A \longrightarrow B$$

and simultaneously:

$$B \longrightarrow A$$

Full-duplex communication is particularly useful for interactive applications.

Examples include:

- Telephone calls
- Video conferencing
- Modern switched Ethernet communication
- Interactive network applications

14 Comparison of Communication Modes

Mode	Direction	Example
Simplex	One direction only	Broadcast communication
Half-Duplex	Both directions, but one at a time	Walkie-talkie
Full-Duplex	Both directions simultaneously	Telephone communication

A simple way to remember them is:

- **Simplex:** One-way
- **Half-duplex:** Two-way, but take turns
- **Full-duplex:** Two-way at the same time

These transmission modes describe how information moves between communicating devices.

15 Putting Everything Together

Let us consider a simple example.

Suppose a student uses a laptop to open a website.

The laptop is the initial sender.

The student enters a website address into the browser.

The browser generates a request.

The request is processed according to networking protocols.

The laptop sends the information through a communication medium.

If the laptop is connected through Ethernet, the communication is wired.

If the laptop is connected through Wi-Fi, the access link is wireless.

The data travels through one or more network devices.

Routers forward packets toward the destination.

Eventually, the web server receives the request.

The server processes it and sends a response.

Now the direction of communication is reversed:

Web Server $\longrightarrow$ Internet $\longrightarrow$ Router $\longrightarrow$ Laptop

Therefore, during normal Internet use, communication is generally two-way.

Many different networking functions may be involved, including:

- Addressing
- Routing
- Forwarding
- Error detection
- Flow control

- Multiplexing
- Demultiplexing
- Segmentation
- Reassembly
- Security

This demonstrates why a computer network is much more than simply connecting two computers with a cable.

16 Important Concept: Network Functions Are Distributed

Students should remember that all network functions are not performed by one single device.

Different functions may be performed at different layers and by different protocols.

For example, the OSI reference model divides network communication into seven conceptual layers.

The layers are:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

The OSI model is a conceptual framework used to understand how different networking responsibilities are organized.

For example:

- The physical layer deals with physical transmission of bits.
- The data link layer deals with frames and local delivery.
- The network layer deals with logical addressing and routing.

- The transport layer provides end-to-end communication functions such as segmentation, flow control, and, depending on the protocol, reliability.
- Upper layers support application-level communication and data representation.

The OSI model will be studied in detail in later lectures.

17 Key Differences: Data, Signal, Medium and Protocol

These four concepts should not be confused.

17.1 Data

Data is the information that the sender wants to communicate.

Example:

Hello Kamran

or a photograph, video, document, etc.

17.2 Signal

A signal is the physical representation used to carry information.

It may involve:

- Electrical signals
- Optical signals
- Radio-frequency electromagnetic signals

17.3 Medium

The medium is the path used by the signal.

Examples:

- Copper cable
- Fiber-optic cable
- Air/free space for wireless communication

17.4 Protocol

A protocol is the set of rules that controls how communication takes place.

Therefore:

Data → Signal → Medium → Receiver

while the entire communication process is governed by:

Protocols

18 Lecture Summary

A computer network is a system of interconnected devices that communicate and share information or resources.

The fundamental elements of network communication include:

1. Sender
2. Receiver
3. Data
4. Communication medium
5. Protocol

Communication media can be broadly classified as:

- Wired
- Wireless

Computer networks can also be classified according to geographical coverage, including:

- PAN
- LAN
- MAN
- WAN

Important network functions include:

- Error control

- Flow control
- Multiplexing
- Demultiplexing
- Addressing
- Routing
- Forwarding
- Segmentation
- Reassembly
- Synchronization

Additional service-enhancing functions may include:

- Encryption
- Authentication
- Checkpointing and recovery
- Compression
- Quality of Service

Finally, communication can be classified according to direction:

- Simplex
- Half-duplex
- Full-duplex

The most important idea to remember is:

A computer network is not merely a physical connection between computers. It is a complete communication system involving devices, data, signals, communication media, protocols, addressing, control mechanisms, and many other functions.

19 Important Points to Remember

1. A network connects two or more devices for communication and resource sharing.
2. The sender originates data.
3. The receiver receives data.
4. A communication medium provides the path for transmission.
5. Communication can be wired or wireless.
6. Protocols define the rules of communication.
7. Computer data is normally represented digitally using bits.
8. Wireless communication can carry digital data.
9. A wireless physical signal is an electromagnetic waveform used to represent and carry the digital information.
10. Error control helps detect and recover from transmission problems.
11. Flow control prevents a fast sender from overwhelming a slower receiver.
12. Multiplexing allows multiple data streams to share a communication resource.
13. Demultiplexing separates received traffic and delivers it to the appropriate destination or application.
14. Addressing helps identify destinations and communication endpoints.
15. Routing determines paths through interconnected networks.
16. Simplex communication is one-way.
17. Half-duplex communication is two-way but not simultaneous.
18. Full-duplex communication is two-way and simultaneous.
19. Encryption protects information from unauthorized access.
20. Checkpointing can help systems resume from a previously saved point after interruption.

20 Review Questions

1. Define a computer network in your own words.
2. What are the five basic elements of computer communication?
3. What is the role of a sender?
4. What is the role of a receiver?

5. What is a communication medium?
6. Differentiate between wired and wireless communication.
7. What is a communication protocol?
8. Why are protocols necessary in computer networks?
9. What is the difference between digital data and a physical signal?
10. Is wireless communication necessarily analog? Explain.
11. What is PAN?
12. What is LAN?
13. What is MAN?
14. What is WAN?
15. Define error control.
16. What is flow control and why is it necessary?
17. Explain multiplexing with an example.
18. Explain demultiplexing with an example.
19. What is network addressing?
20. What is routing?
21. What is segmentation?
22. What is encryption?
23. What is checkpointing?
24. Define simplex communication.
25. Define half-duplex communication.
26. Define full-duplex communication.
27. Give one example of each transmission mode.
28. What is the difference between half-duplex and full-duplex communication?
29. Why are network functions distributed across different layers?

21 Short Classroom Activity

Consider the following situation:

A student uses a laptop connected to Wi-Fi to download a PDF file from a university server.

Identify the following:

1. Who is the sender at the beginning of the request?
2. Who is the receiver?
3. What is the data?
4. What is the communication medium?
5. Is the access connection wired or wireless?
6. Which protocols may be involved?
7. Why is addressing required?
8. Why is error control useful?
9. Why is flow control useful?
10. Can multiple applications use the same network connection?
11. If yes, which concept allows this?
12. Is communication between the laptop and server only one-way?
13. Which communication mode describes simultaneous two-way communication?

22 References

References

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